

Universitatea Tehnica Cluj-Napoca
Electronica Telecomunicatii si Tehnologia Informatiei

Proiect CAD
VU-metru

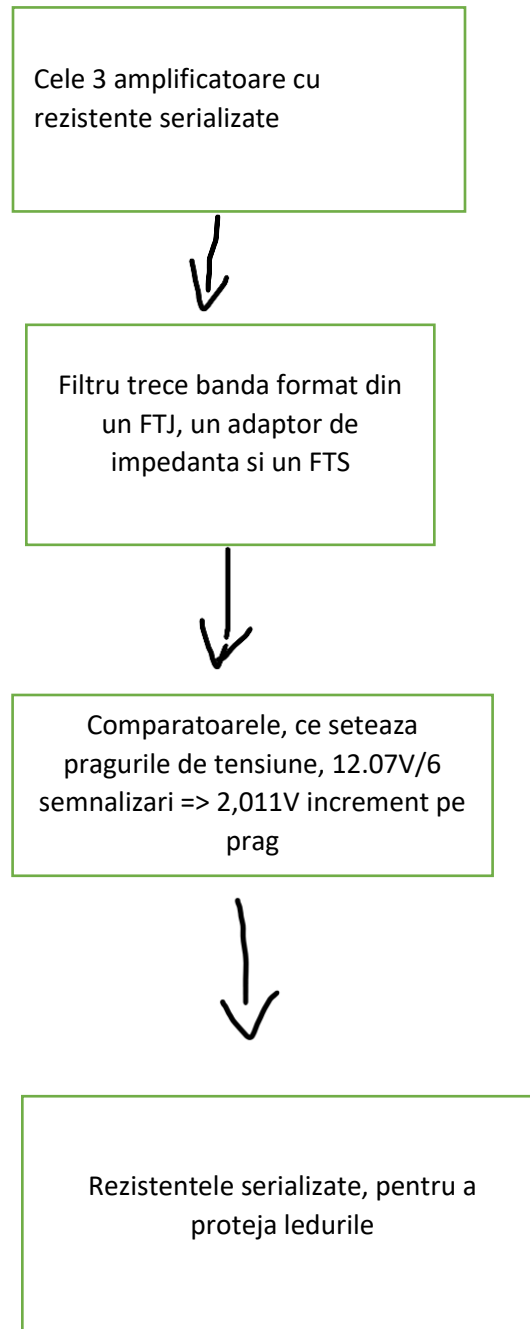
Dobrea Marius Dorian

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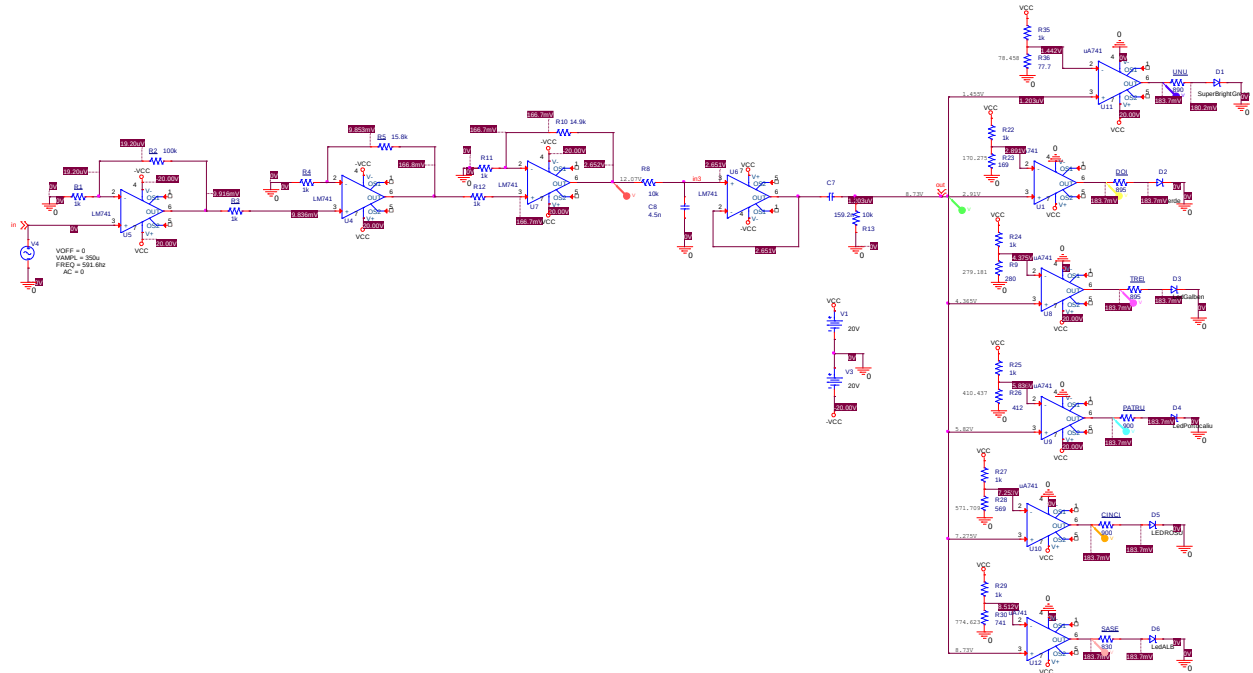
Sa se proiecteze un circuit electronic care sa indice pe un afisaj cu LED-uri nivelul semnalului audio dintr-o banda de frecventa specificata (VU-metru). Circuitul este alimentat de la tensiunea $\pm VCC$. LED-urile trebuie sa fie de culori diferite pentru fiecare domeniu specificat:

- Amplitudinea semnalului de intrare [uV]:350uV
- Banda de frecventa [Hz]: $f_{min}= 100\text{Hz}$ si $f_{max}=3500\text{Hz}$
- $VCC=20$
- Semnalizari=6

Schema bloc:



Schema elettrica:



Blocul de amplificare :

Pentru amplificator avem o amplificare de: $A=1+R_2/R_1$

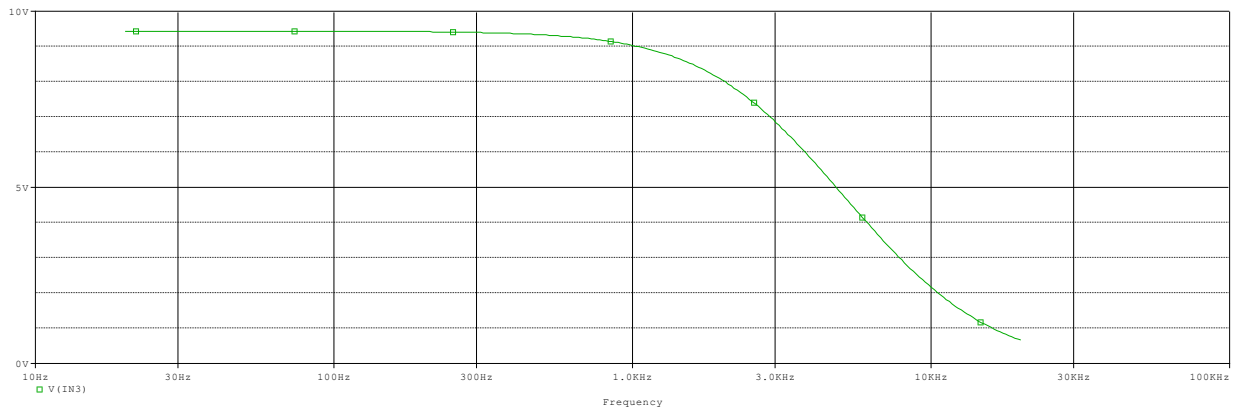
$$A1 = 1 + 100k/1k = 101 \text{ ori} \Rightarrow 45.1 \text{mv}$$

$$A_2 = 1 + 15.8k/1k = 16.8 \text{ ori} \Rightarrow 759.2\text{mv}$$

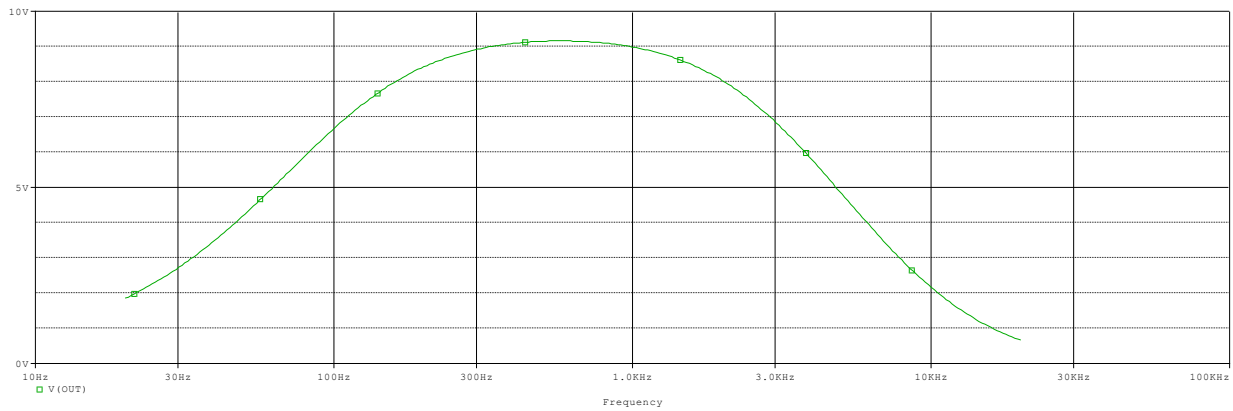
$$A3 = 1 + 14.9K/1K = 15.9 \text{ ori} \Rightarrow 12.7V$$

ACSweep:

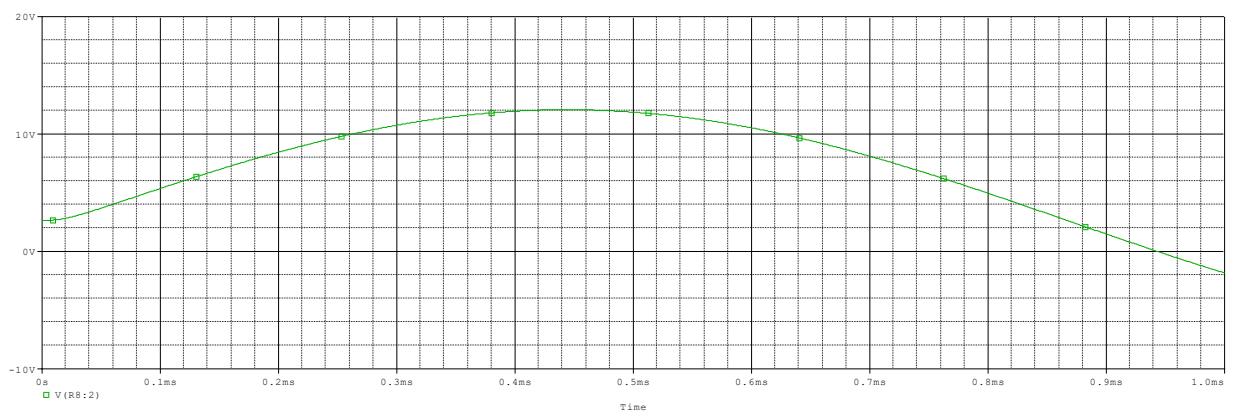
FTJ:



FTJ+FTS



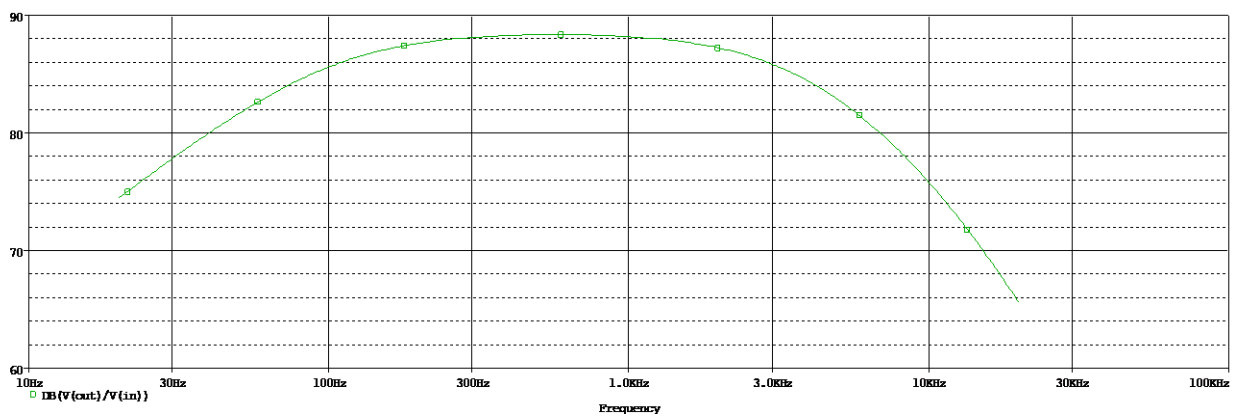
Time domain



INAINTE DE FILTRU → Amplitudinea maxima=12.07v

DUPA DE FILTRU → Amplitudinea maxima=12.07v

Filtru trece banda :



Trace Color	Trace Name	Y1	Y2	Y1 - Y2	Y1(Cursor1) - Y2(Cursor2)	497.948m			
	X Values	100.000	3.5050K	-3.4050K	Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
CURSOR 1.2	DB(V(out)/V(in))	85.600	85.103	497.948m	0.000	0.000	85.600	85.103	85.352

Adaptorul de impedanta se comporta ca si un buffer intre filtrul trece sus si filtrul trece jos, pentru a le putea calcula frecventa de taiere separa (fara a interfera unul cu celalalt).

Frecventa de taiere: $F_c = 1/2\pi RC$

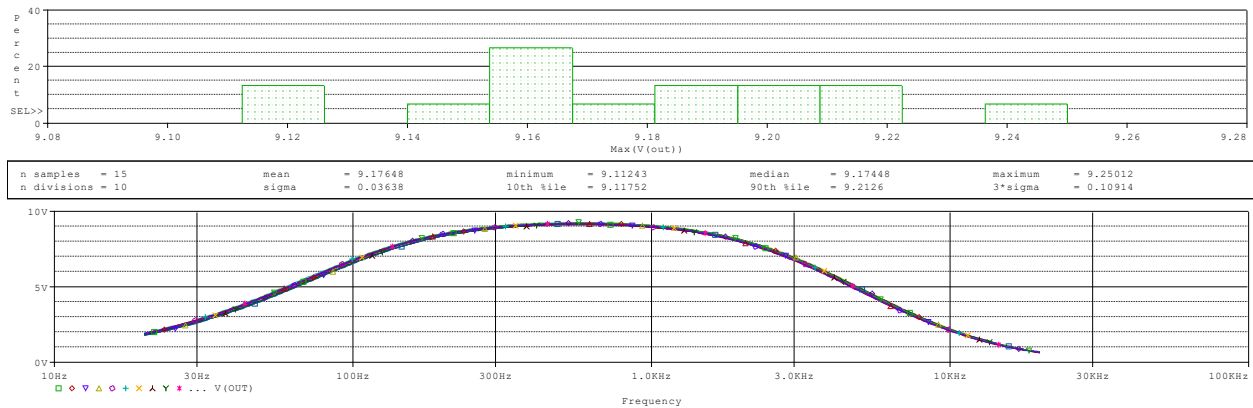
Am ales $R = 10k$ iar apoi am calculate condensatoarele

$$F_{c1} = 3.5kHz = 1/2 * \pi * 10k * 4.5n$$

$$F_{c2} = 100Hz = 1/2 * \pi * 10k * 159.2n$$

$$\text{Frecventa centrala} = f_r = \sqrt{F_{c1} * F_{c2}} = 591.6Hz$$

Analiza de tip Monte Carlo:



Comparatoare:

Deoarece sunt 6 semnalizari (0-8.73V)

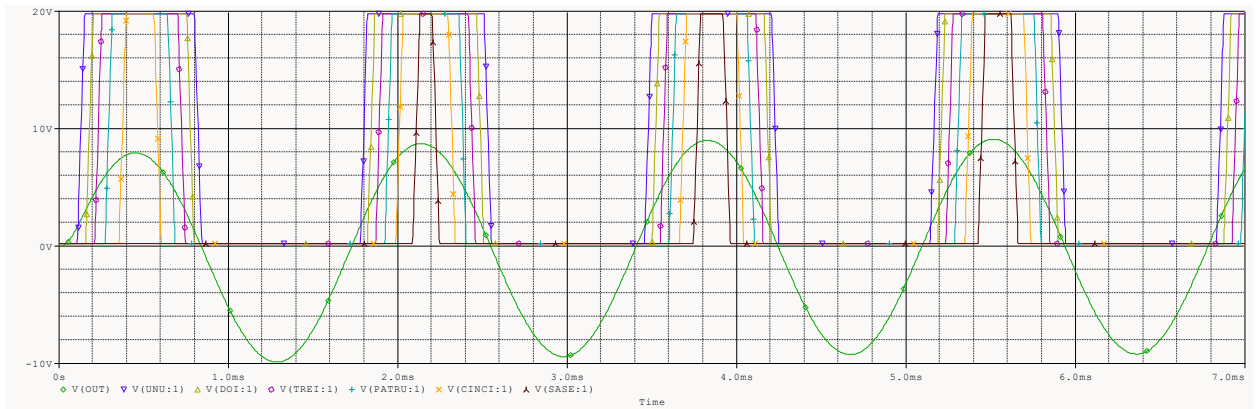
Praguri de comparare:

- 1.455V
- 2.91V
- 4.365V
- 5.82V
- 7.275V
- 8.73V

$$\text{Tensiune iesire} = (\text{tensiune intrare} \times R_2) / (R_1 + R_2) \Rightarrow R_2$$

Am ales $R_1 = 1k$ iar R_2 corenspunzator pentru fiecare prag de mai sus :

- 77.7 ohm
- 169 ohm
- 280 ohm
- 412 ohm
- 569 ohm
- 741 ohm

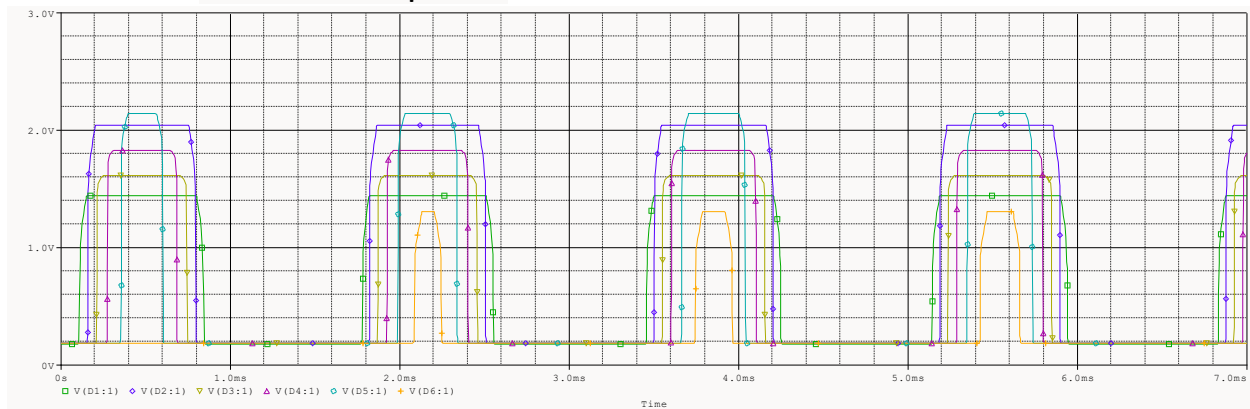


Rezistenta dintre comparatoare si leduri:

$$R = (V_{CC} - V_{led}) / I_{led}$$

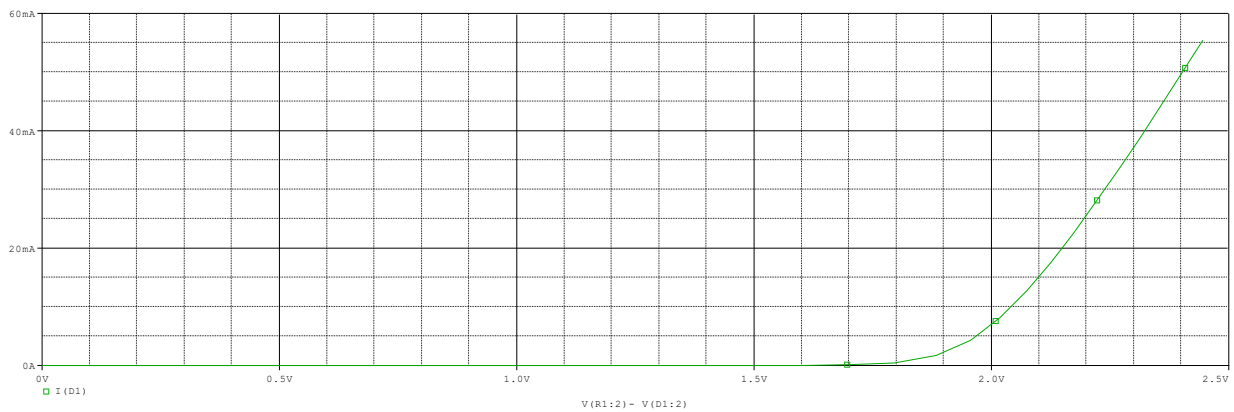
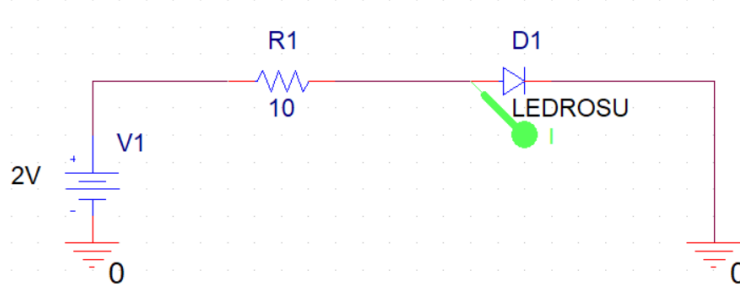
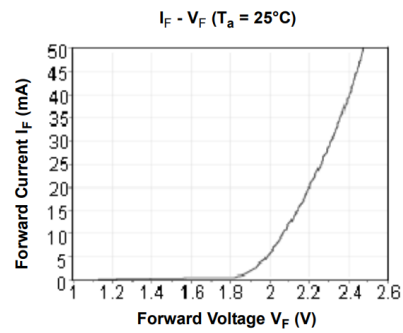
- 1) $R = (20V - 2.2V) / 20mA = 890 \text{ ohm}$ -> Super Bright Green
- 2) $R = (20 - 2,1) / 20mA = 895 \text{ ohm}$ -> Verde
- 3) $R = (20 - 2,1) / 20mA = 895 \text{ ohm}$ -> Galben
- 4) $R = (20V - 2V) / 20mA = 900 \text{ ohm}$ -> Portocaliu
- 5) $R = (20V - 2V) / 20mA = 900 \text{ ohm}$ -> Rosu
- 6) $R = (20V - 3.4V) / 20mA = 830 \text{ hm}$ -> Alb

Simulare Tensiune pe leduri

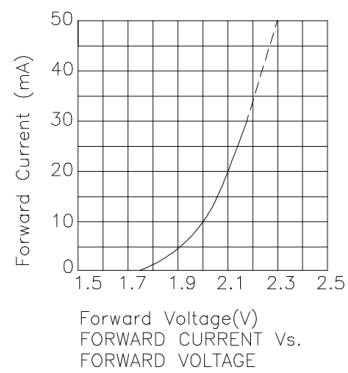
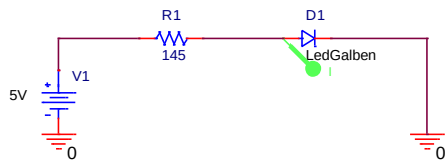


1)

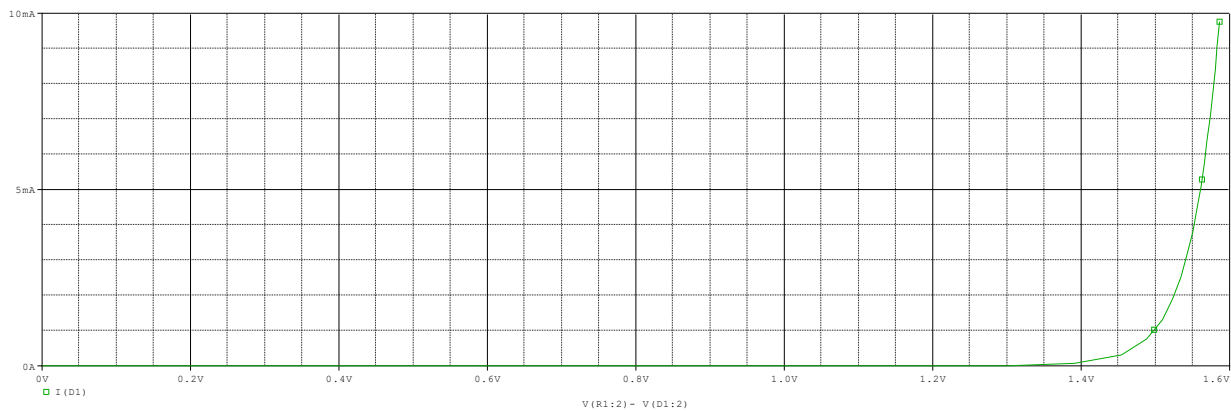
Typical Characteristics



2) Led Galben



$$R=145=(5V-2.1V)/20mA$$



3) Led Verde

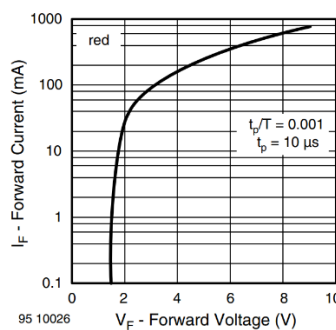
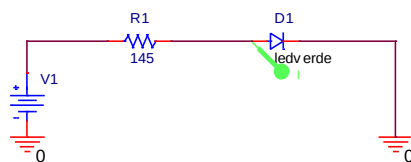
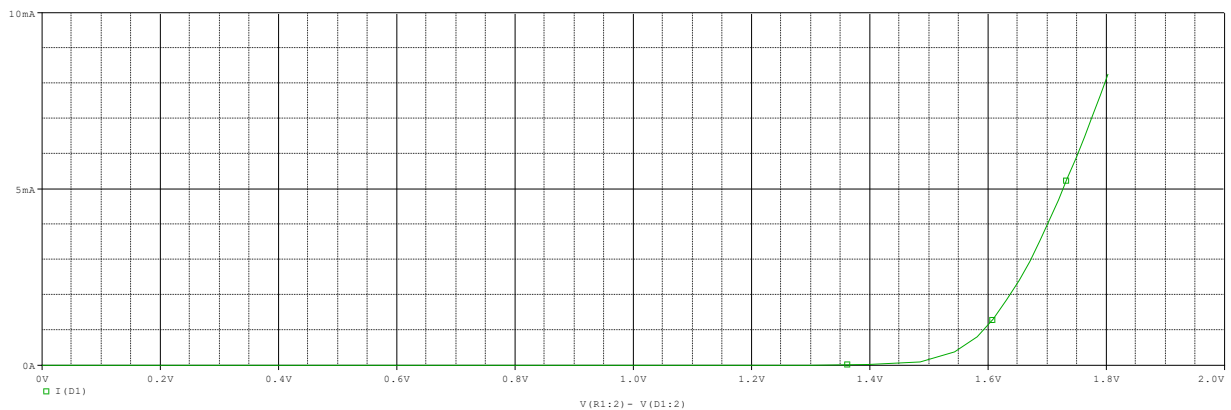


Fig. 4 - Forward Current vs. Forward Voltage



$$R=145=(5V-2.1V)/20mA$$



4) Led Portocaliu

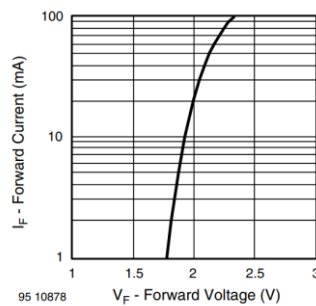
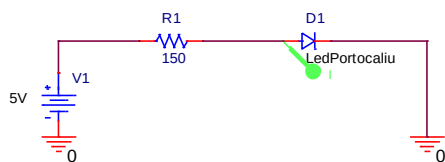
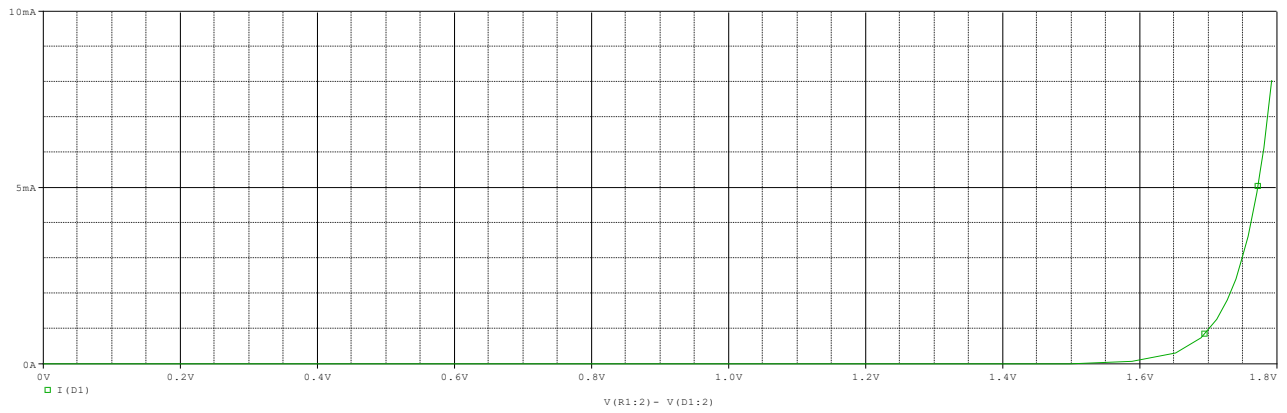
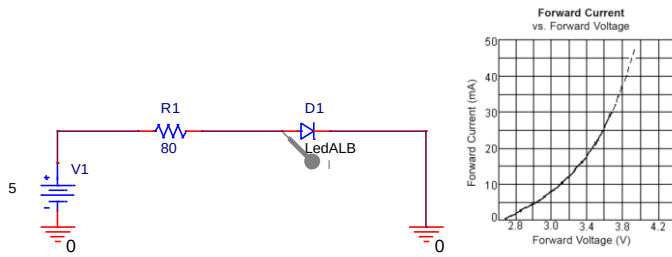


Fig. 3 - Forward Current vs. Forward Voltage

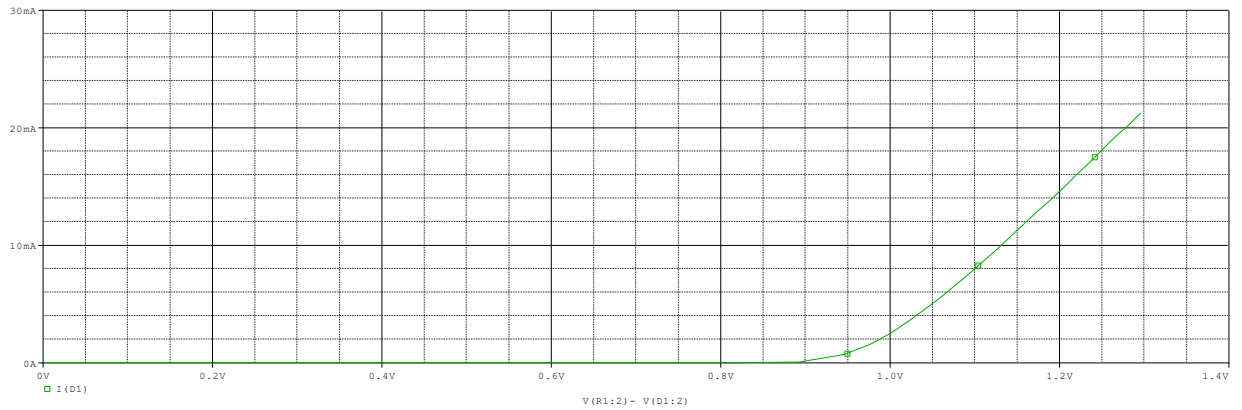
$$R=145=(5V-2.1V)/20mA$$



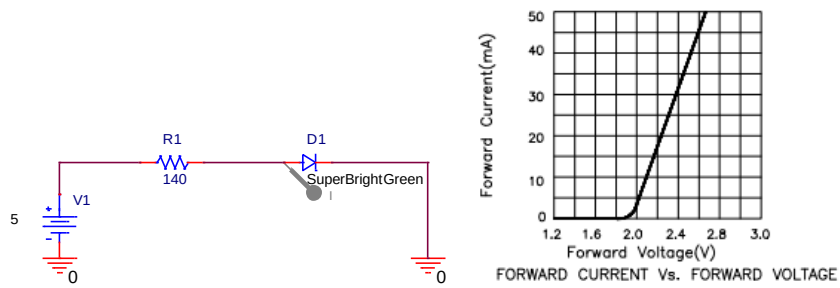
5) Led Alb



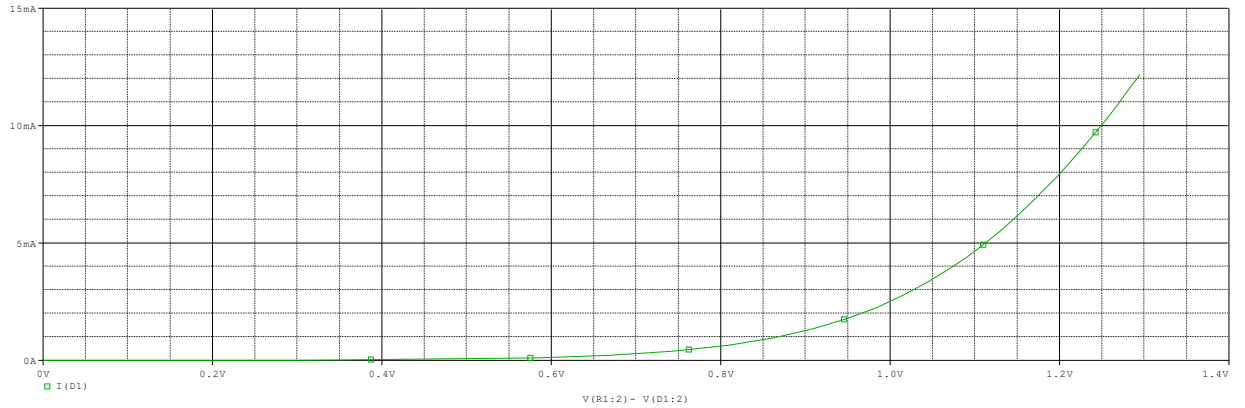
$$R = (5V - 3.4V) / 20mA = 80 \text{ ohm}$$



6) Led Ultra Bright Green



$$R = (5V - 2.2V) / 20mA = 140 \text{ ohm}$$



Datasheets:

- Led Rosu: <https://www.farnell.com/datasheets/1498852.pdf>
- Led Galben: <https://www.arduino.cc/documents/datasheets/LEDY-L-7113YT.pdf>
- Led Verde: <https://www.farnell.com/datasheets/1671521.pdf>
- Led Portocaliu: <https://www.vishay.com/leds/list/product-83134/>
- Led Ultra Bright Green: https://www.digchip.com/datasheets/parts/datasheet/000/L-53SGCDMP0_0579-pdf.php
- Led Alb : <https://www.farnell.com/datasheets/2861562.pdf>